Laboratory 12

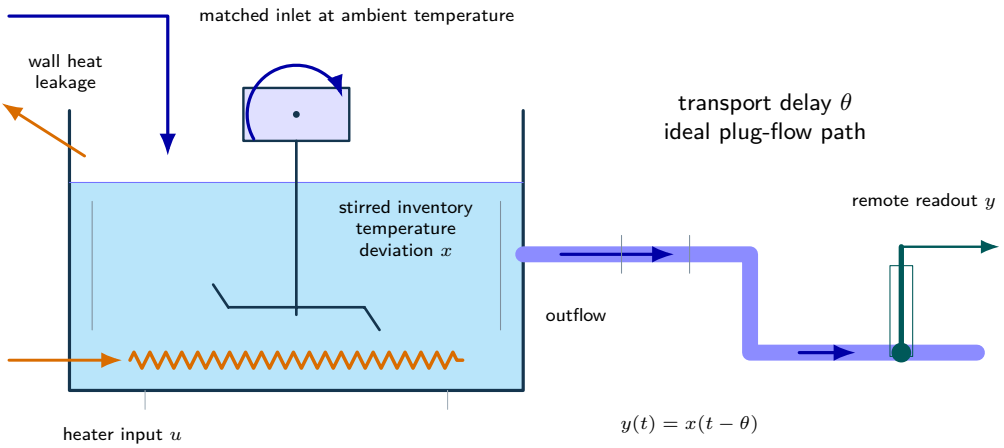
A synthetic process from identification to validation

Exercise sheet – five integrated problems

Evidence boundary. Original hypothetical scenarios. Derive, calculate and justify decisions; every numerical record is synthetic. Figures are conceptual and not to scale. **Shared nominal data.** Synthetic process: $G(s) = 1.8e^{-0.6s}/(4s+1)$. Identification and validation use separate generated records. No parameter estimate or fitted response is a physical measurement.

1. Inferring an aggregate thermal process from a step

The illustration shows a well-mixed thermal inventory followed by an ideal transport delay. The drawing supplies a physical interpretation, while the declared parameters define a synthetic teaching model.



One physical interpretation of a first-order-plus-delay model

Synthetic experiment
$G(s) = 1.8e^{-0.6s}/(4s+1)$. Declared gain, lag and delay generate the records; none is a physical measurement.

Thermal interpretation
One mixed energy store plus ideal transport. Through-flow and wall losses enter one effective conductance.

Identification boundary
Fit on training data, freeze the model, then check a separate multistep record. Geometry does not identify capacity or heater efficiency.

A stirred thermal inventory followed by ideal transport is one possible physical interpretation of the identified transfer function. The effective conductance includes wall leakage and through-flow heat removal. Its fitted gain and time constant do not separately determine capacity, conductance and heater efficiency.

(a) A synthetic process is $G(s) = Ke^{-\theta s}/(\tau s+1)$ with $K = 1.8$, $\tau = 4$ s and $\theta = 0.6$ s. A 2-unit step starts at $t_0 = 2$ s. Derive the zero-state output and find its final value and the time at which it reaches 63.212 percent of that final value.

(b) In a deliberately noise-free synthetic step record with amplitude 2, the final output is 3.6, onset occurs 0.6 s after input changes, and the 63.212 percent point occurs 4 s after onset. Recover K, delay and time constant. Explain two obstacles with noisy sampled data.

(c) For $C\dot{x} = \eta u - Hx$ followed by pure delay, relate K, τ to C, η, H. Show that $K = 1.8$, $\tau = 4$ do not uniquely identify all three physical parameters.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

2. Fitting a delayed model without leaking validation information

Fit only the training record, using an admissible parameterization. Keep independent excitation and noise for validation, and examine how the numerical delay representation affects optimization.

(a) Write a least-squares objective for fitting K, time constant and delay to sampled step data. Propose a positive-parameter reparameterization and explain its limitation when zero delay is a plausible model.

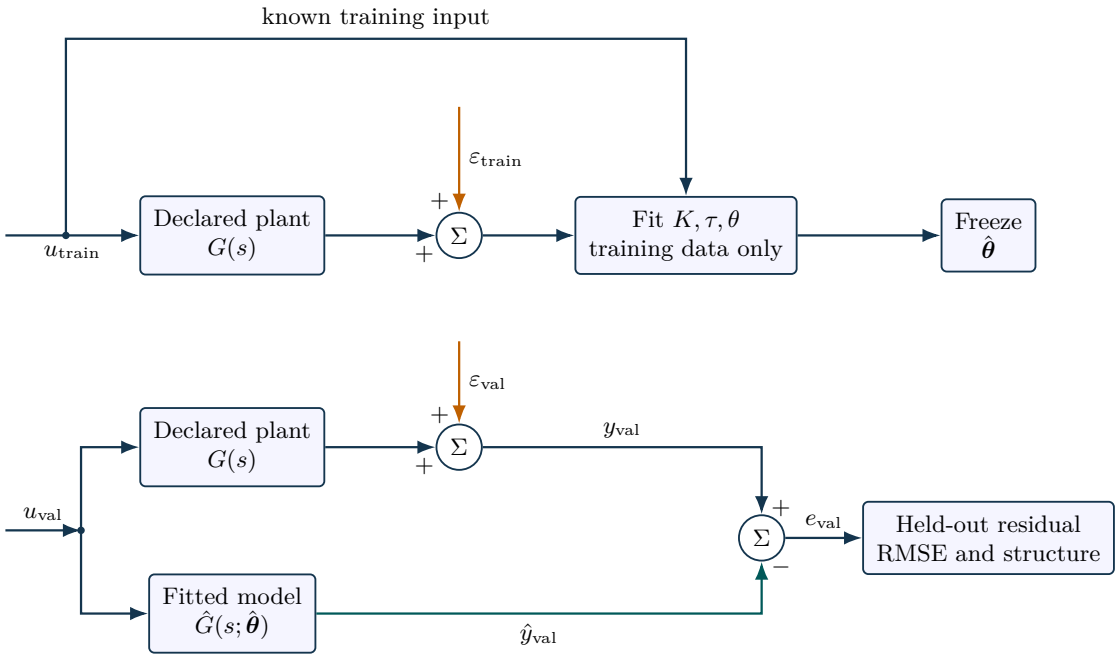
(b) Explain why reporting only training RMSE after fitting a noisy step response gives incomplete evidence. Specify a held-out synthetic input record and noise-generation procedure that avoids reusing the training noise.

(c) What happens if a gradient-based fit represents delay only by a buffer rounded to an integer number of fixed time steps? Propose a more informative fitting representation and a subsequent implementation check.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

3. A held-out multistep test with a statistical reference

The validation record uses new input transitions and independent noise. Interpret residual magnitude using a specified stochastic model rather than claiming a perfect model from one small RMSE.



Different input record and independent noise; no refitting on validation data.
All records in this exercise are synthetic.

Training selects parameters; held-out validation freezes them. Separate noise records avoid reusing training residuals as validation evidence.

(a) Use input increments $+1$ at 1 s, -0.6 at 8 s, $+1.1$ at 15 s and -0.8 at 22 s. Write the exact synthetic model output and find its eventual equilibrium.

(b) Validation adds independent zero-mean Gaussian noise of standard deviation 0.015. Explain the expected mean squared residual for a perfect deterministic model. Does a realized RMSE of 0.0145 prove the identified dynamics are exact?

(c) For a perfect fixed deterministic model and $N = 400$ independent Gaussian validation errors with $\sigma = 0.015$, derive a central 95 percent reference interval for realized RMSE. Explain why it is not a confidence interval proving the fitted dynamics correct.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

4. Tuning a PI law subject to uncertain steady capability

Treat the tuning rule as a proposal and separately test whether the actuator can maintain the requested operating point under gain uncertainty and a constant disturbance.

(a) Use $k_p = \tau/[K(\lambda + \theta)]$, $k_i = k_p/\tau$ with $\lambda = 2$ s and nominal parameters. Calculate both gains and their changes if the assumed delay rises to 0.9 s.

(b) The process input is constrained to $[0, 2]$, and the reference is one. A constant disturbance of -0.2 is added inside the first-order state equation as $\tau\dot{y} = -y + Ku(t - \theta) - 0.2$. Find the required equilibrium input before and after the disturbance.

(c) Suppose the true gain lies in $[1.5, 2.1]$, $u \in [0, 2]$, and the disturbance in the state equation is -0.2. Find the largest reference guaranteed feasible at equilibrium for all allowed gains. Can increasing λ make a reference of 3 feasible?

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

5. Implementing a noninteger transport delay

The nominal simulation uses an exactly aligned delay, while a fitted delay need not align with the numerical grid. Preserve this distinction in uncertainty tests and reproduction notes.

(a) Let $h = 0.01$ s and delay 0.6 s, with zero-based indexing. Derive the exact held-input plant update and state which stored control sample is used at index k. Specify the first index at which an initial command u_0 influences the update.

(b) The fitted parameters are approximately $[1.800025, 3.993491, 0.604289]$. Describe how to evaluate delay and gain uncertainty without claiming physical validation. Include training/validation separation, actuator limits and a numerical convergence check.

(c) For $\theta = qh + \delta$ with integer q and $0 \leq \delta < h$, derive the exact held-input update of the first-order process over one step. Identify both historical input samples used and apply the indexing rule to $\theta = 0.604289$ s, $h = 0.01$ s.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.